

Corporate Access & Intelligence System

Total Website Blueprint, System Architecture & Implementation Plan

1. Executive Overview & System Purpose

The **Corporate Access & Intelligence System** is a high-performance, enterprise-grade data analytics and workforce intelligence application designed to process multi-million row datasets directly in the browser with zero server-side latency. The platform delivers instant visual charting, dynamic multi-dimensional filtering, statistical data profiling, anomaly detection, and an integrated **Personal Profile & Login Activity System**.

Core Value Proposition: Sub-second multi-million row processing via Web Workers, rich glassmorphism UI with ambient aurora background, real-time localized session tracking, and full user profile customization.

2. Complete Technical Architecture

The application is built using modern web standards to ensure high reliability, visual elegance, and client-side privacy:

Layer / Component	Technology Stack	Key Functional Responsibility
Frontend Framework	React 18.2 + Vite 5	Declarative component rendering and fast HMR development environment.
Styling System	Vanilla CSS3 & CSS Variables	Custom design system with root rem scaling, dark/light themes, aurora mesh background.
Data Engine	Dedicated Web Worker (PapaParse)	Off-main-thread non-blocking parsing of multi-million row CSV files.
Data Visualization	Chart.js 4 + react-chartjs-2	Dynamic Bar, Doughnut, Line, Pie, Radar, and Polar Area interactive charts.
Authentication & Session	Local Storage & Activity Tracker	Automated, read-only login/logout tracking with local timezone enforcement.
Iconography	Lucide React Icons	Clean modern UI iconography across navigation, cards, tables, and modals.

3. Personal Profile & Login Activity System Specification

A cornerstone feature of the platform is the full-featured, secure user profile center accessible by clicking the user profile badge at the top right of the application header.

Left Side: Personal Profile Card

- **Profile Avatar & Photo Picker:** Prominent enlarged avatar circle (115px × 115px) with Camera overlay button. Supports preset executive avatars, custom image file uploads (PNG, JPG, WEBP), and initial badges.
- **Personal Information:** Full Name, Verified Corporate Checkmark, Email Address, Mobile Number, Role/Title, Department, Account Creation Date.
- **Skills & Interests:** Interactive skill chips (Data Analytics, Workforce Intelligence, React.js, Python, Financial Modeling, Predictive AI).
- **Inline Edit Form:** Allows instant updating and saving of profile metadata stored locally.

Right Side: Login & Logout Tracking Dashboard

- **Live Active Session Ticker:** Real-time updating counter ticker (seconds precision) displaying current active login duration.
- **5 Metric Dashboard Cards:**
 1. **LAST LOGIN:** Exact localized timestamp of most recent sign-in.
 2. **CURRENT SESSION:** Real-time updating session duration ticker.
 3. **TOTAL LOGIN HOURS:** Cumulative total time spent across all sessions.
 4. **LOGIN COUNT:** Total count of recorded login sessions.
 5. **LAST LOGOUT:** Exact localized timestamp of previous logout or active state.
- **Automated Login History Table:** Read-only, tamper-proof session table recording Date, Login Time, Logout Time, Duration, Device/Location, and Status (Active / Logged Out) formatted in user's local timezone.
- **Full Screen Toggle:** Dedicated Maximize/Minimize button allowing edge-to-edge full screen modal view.

4. Core Data Analytics Modules & Visual Studio

The platform provides an end-to-end data analytics suite organized into logical tabs and metric folder decks:

Module Name	Key Features & Capabilities
Interactive Sidebar Filters	Global text keyword search, categorical checkboxes with row counts, dynamic numeric range sliders, preset risk level selectors (Low, Medium, High, All).
Metrics Directory Folders	Structured folder stack: Quality & Health Folder (Missing values, duplicates, completeness score), Financial Folder , Dataset & Filters Folder .
Executive Dashboard View	High-impact visual analytics deck featuring Department workforce distribution bar chart and Work Mode breakdown doughnut chart.
Custom Visual Studio	Custom chart builder allowing users to choose X-axis, Y-axis, aggregation types (Sum, Mean, Count), and chart styles.
Data Explorer Table	High-performance paginated table supporting multi-column sorting, live row filtering, and instant CSV export.
Statistical Profiling	Detailed metrics summary calculating Min, Max, Mean, Median, Standard Deviation, and Null Ratios for every column.
Comparison Analysis	Side-by-side comparative analytics matrix for workforce segments, department metrics, and historical runs.

5. Visual Design System & Aesthetics

The user interface follows modern luxury web design standards:

- **Aurora Mesh Background:** Atmospheric radial gradients blending subtle indigo, purple, cyan, emerald, and rose light glows.
- **Glassmorphism:** Card elements use backdrop blur (backdrop-filter) and translucent linear gradient borders.
- **Dynamic Theme Modes:** Seamless switching between Dark Mode (#090d16) and Light Mode (#f6f8fd).
- **Fluid Typography:** Base font size scaled to 110% (`17.6px`) with Google Fonts (Outfit & Inter) for maximum legibility.

6. Future Expansion Roadmap

1. **AI Insights Copilot:** Automated natural language dataset summaries and anomaly explanations.
2. **Real-Time WebSockets:** Streaming data ingestion for live enterprise workforce event tracking.
3. **Multi-Tenant RBAC:** Granular role-based permissions (Viewer, Analyst, Admin) and team workspace sharing.

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